RCA-2025-007 Unit 2 draft tube cone cracks and runner cavitation

Level 2 root cause analysis

Document	RCA-2025-007
Revision	Final
Date	2025-12-19
Applies to	CCD-U2-TUR-DT, CCD-U2-TUR-RUN, CH-FR-1987-0043
Owner	Lead: Priya Nandakumar. Team: Tom Halvorsen, Ray Okafor, Dale Whitaker, Karen Doyle (System Ops), Lars Bergström (Cromley Hydraulics)
Approved	Elena Marsh, 2026-01-08
Status	Actions open (A4, A5)

1 Problem statement

At the annual inspection on 20 October 2025, three cracks were found at stiffener welds of the Unit 2 draft tube cone, and cavitation erosion on the runner band. None was present at the 2024 inspection.

2 Impact

Repair outage 216 h after the inspection (WO-2025-0821, WO-2025-0834); approx. 7,400 MWh lost generation; repair cost approx. USD 185k. Risk avoided: a growing cone crack can lead to a draft tube liner failure and a long outage.

3 How it was detected

Visual inspection and NDT at the annual inspection. No vibration or pulsation monitoring existed on the units in 2025. Operators had noted "rough running" and noise on Unit 2 several times in July and August 2025 (shift log), without follow-up.

4 Timeline

Date / time	Event	Source
26 May 2025	MOC-2025-012 raised by System Ops: bypass the rough-zone table so units can hold 15–35 MW for a regulation reserve contract	MOC register
30 May 2025	MOC approved by the plant manager; reliability review not requested	MOC register
2 Jun 2025	Bypass enabled by I&C	WO history
Jun–Sep 2025	Units regularly held at part load. Unit 2 was the preferred regulating unit (lowest start count). Estimated time in the rough zone from plant controller logs: **Unit 2 approx. 640 h**, Unit 1 approx. 210 h, Unit 3 approx. 180 h	Plant controller logs
Jul–Aug 2025	Shift log notes "U2 rough running / noisy" on 5 occasions	Shift log
30 Sep 2025	MOC expiry date passed; bypass not removed	MOC register
20 Oct 2025	Annual inspection: cone cracks 80, 140, 210 mm at stiffener weld toes; runner band cavitation over 0.4 m^2, max. depth 4 mm	WO-2025-0821
25 Oct–3 Nov 2025	Weld repair (NDT ok) and runner band repair with CH field service	WO-2025-0834
8 Dec 2025	Bypass removed, rough-zone table restored	WO history

5 Evidence collected

Parts: crack replicas and photos, cavitation depth map. Position: crack locations in the cone zone where the vortex rope pressure pulsation acts. People: shift supervisors, System Ops, I&C. Paper: plant controller setpoint logs (used to estimate hours in the zone), MOC records, shift log, 2014 overhaul weld records, CH assessment by Lars

Bergström. Paradigms: "the units have always run at part load sometimes, it's fine".

6 Hypotheses and verification

#	Hypothesis	Evidence	Result
H1	Fatigue from draft-tube vortex rope pulsation (part-load operation)	640 h in the rough zone for Unit 2 vs. CH guidance of 50 h/year; crack location; CH assessment	**Verified**
H2	Poor weld toe profile from the 2014 overhaul	Weld toe angle steep at the cracked stiffeners	Verified as **contributing**
H3	Cavitation from low tailwater	Tailwater normal all summer	Ruled out
H4	Cavitation from high load / high head	Unit 2 rarely above 90 % gate in the period	Ruled out
H5	Cavitation from part-load operation	Erosion pattern at the band typical for part-load inter-blade vortices; coincides with H1	**Verified**

7 Root causes

Type	Root cause	Evidence
Physical	Fatigue cracking of the cone stiffener welds and runner band cavitation from sustained operation in the vortex-rope zone (contributing: steep weld toes)	Inspection, hours estimate, CH assessment
Human	Bypass enabled and left in place beyond its expiry; Unit 2 used as the preferred regulating unit; "rough running" reports not escalated	MOC register, shift log
Latent	MOC process allowed a rough-zone bypass without reliability review against the OEM limit (CH-DOC-OM-0101 section 3)	MOC-2025-012
Latent	Commercial value of the regulation reserve was weighed without the equipment cost; no agreed band outside the rough zone	Interview System Ops
Latent	No tracking of rough-zone hours and no vibration or pulsation monitoring	Design, KPIs

8 Actions

#	Action	Root	Owner	Due	Status
A1	Remove the bypass, restore the rough-zone table	Physical, Human	Dale Whitaker	2025-12-15	Closed 2025-12-08
A2	OP-03 rev C: reliability engineer sign-off mandatory for any rough-zone bypass; limit on hours; not on units with open findings (MOC-2025-019)	Latent 1	Priya Nandakumar	2026-01-31	Closed 2026-01-15
A3	Weld repair with improved toe profile; runner band repair	Physical	Tom Halvorsen	2025-11-15	Closed 2025-11-03
A4	Implement a rough-zone hours KPI per unit (monthly review) using CH monitoring PLC data	Latent 3	Dale Whitaker / Priya Nandakumar	2026-12-31	**Open**
A5	Commercial: renegotiate the regulation reserve band so it can be delivered outside the rough zone	Latent 2	Karen Doyle	2026-06-30	**Open, overdue**
A6	Inspect Unit 2 draft tube cone at the 2026 annual inspection	Physical	Tom Halvorsen	2026-10-31	Closed 2026-10-22, no new cracks

9 Effectiveness check

Rough-zone hours per unit below 50 h per year (CH guidance), reviewed monthly from 2027 once A4 is in place. No new cone cracks at the 2026 and 2027 inspections.

10 Lessons and FMECA / PM updates

FMECA FM-07 added (vortex rope). Lesson: part-load operation is an operating decision with a maintenance cost; the signature (pulsation near 0.28 × runner frequency, high TGB vibration, MW swings) can now be seen in the CH monitoring data.